

TW05: NumPy Foundations

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Create and inspect NumPy arrays, apply vectorised calculations, select values, use masks and axes, broadcast values, and combine arrays.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Create and inspect NumPy arrays

Scenario

A data analyst needs arrays with predictable shapes and numeric data types.

INSTRUCTIONS

1. Create a one-dimensional array from the supplied temperatures.
2. Create a 2 x 3 array from the supplied nested list.
3. Display ndim, shape, size, and dtype for both arrays.
4. Create arrays of zeros, ones, and a chosen constant.
5. Convert the temperature array to floating-point without changing the original.

REQUIRED SUBMISSION

- Two source arrays and three generated arrays.
- Correct attributes for each source array.
- A converted copy and proof that the original dtype is unchanged.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Replace repeated calculations with vectorisation

Scenario

A shop wants subtotal, discount, tax, and final totals for several products without writing a calculation for each row.

INSTRUCTIONS

1. Create price and quantity arrays.
2. Calculate subtotals using one vectorised expression.
3. Apply a 10 percent discount to subtotals above 50 using np.where.
4. Add 20 percent tax after discount.
5. Compare the vectorised result with one manually calculated row.

REQUIRED SUBMISSION

- Four aligned NumPy result arrays.
- A manual validation check for one row.
- A short explanation of vectorisation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Select values with indexing, slicing, and masks

Scenario

A sensor series contains values that must be selected and checked against a safe range.

INSTRUCTIONS

1. Select the first, last, and fourth readings by index.
2. Select readings 2 through 5 with a slice.
3. Create a Boolean mask for readings below 10 or above 30.
4. Use the mask to extract unusual values and count them.
5. Create a corrected copy where unusual values become np.nan.

REQUIRED SUBMISSION

- Correct indexed and sliced outputs.
- A Boolean mask, extracted values, and count.
- A corrected floating-point copy with the source unchanged.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Summarise a matrix and use broadcasting

Scenario

Rows represent shops and columns represent three months of revenue.

INSTRUCTIONS

1. Calculate overall, per-shop, and per-month totals and means using axis.
2. Add a different fixed bonus to every month using broadcasting.
3. Calculate each shop's difference from the monthly mean.
4. Verify the shapes before and after broadcasting.
5. Explain the difference between axis=0 and axis=1 in this matrix.

REQUIRED SUBMISSION

- Six correctly labelled summaries.
- A broadcast bonus result and centred matrix.
- Shape checks and an axis explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Combine arrays and handle missing values

Scenario

Two branches send compatible data matrices, but one value is missing.

INSTRUCTIONS

1. Stack the two branch matrices vertically.
2. Create a branch-label column and concatenate it horizontally.
3. Replace the missing value with the mean of its column in a copy.
4. Calculate `np.mean` and `np.nanmean` before replacement and explain the difference.
5. Add assertions for final shape and absence of missing values.

REQUIRED SUBMISSION

- Correct vertical and horizontal combinations.
- A justified missing-value replacement.
- Two passing assertions and a mean-versus-nanmean explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____